

Physical Prototype

1. Piano box - length sorting

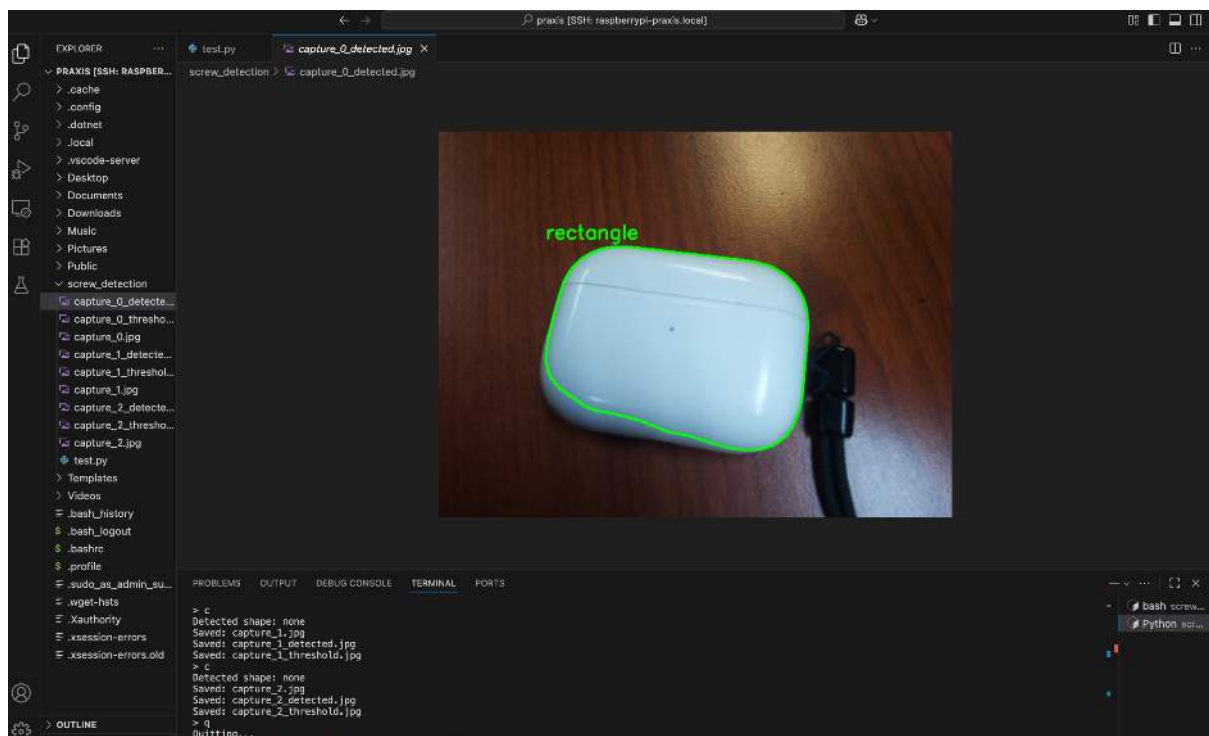


The screws will be transferred horizontally on to the box, with its head tucked into the slot. The box is placed with a 15° angle so the screw can roll down the box by the force of gravity. Drawers are placed inside the box, which allows the screws to fall into corresponding box after sorting

2. Sliding chopsticks - transport screws from one layer to the next



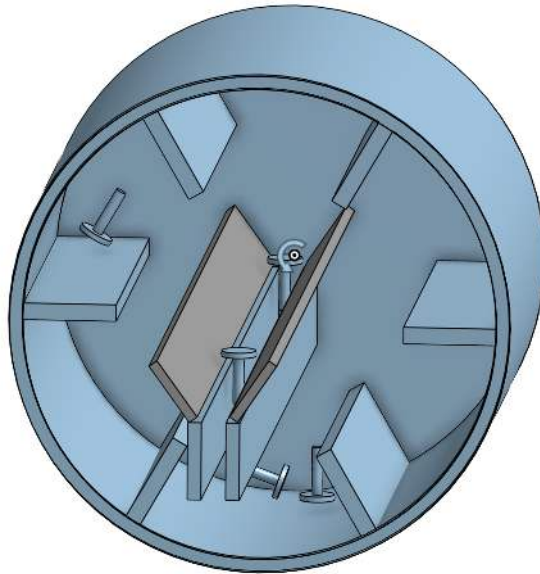
The box is placed with a 30° angle so the screw can roll down between two sticks by the force of gravity. It's designed for the mechanism that require the screw to be placed vertically



First test of raspberry pi 4 + camera module 3 wide lens and test code to detect the shape of objects. This is the first test of the camera, but it shows it is possible to detect shapes even in bad conditions (lighting, distance).

Model Prototype

1. Drum feeder



The drum feeder is formed by a hollow cylinder with several baffles around the drum, it will rotate with an engine. As the drum rotate, screws will be lifted by the baffles, and as it goes to the top part of the rotation, screws will fall down to the middle convey belt due to gravity, its head will be stuck by the gradually thinning barrier, so this mechanism can make a bulk of screws into an ordered line of screws standing vertically.

2. Vibratory bowl

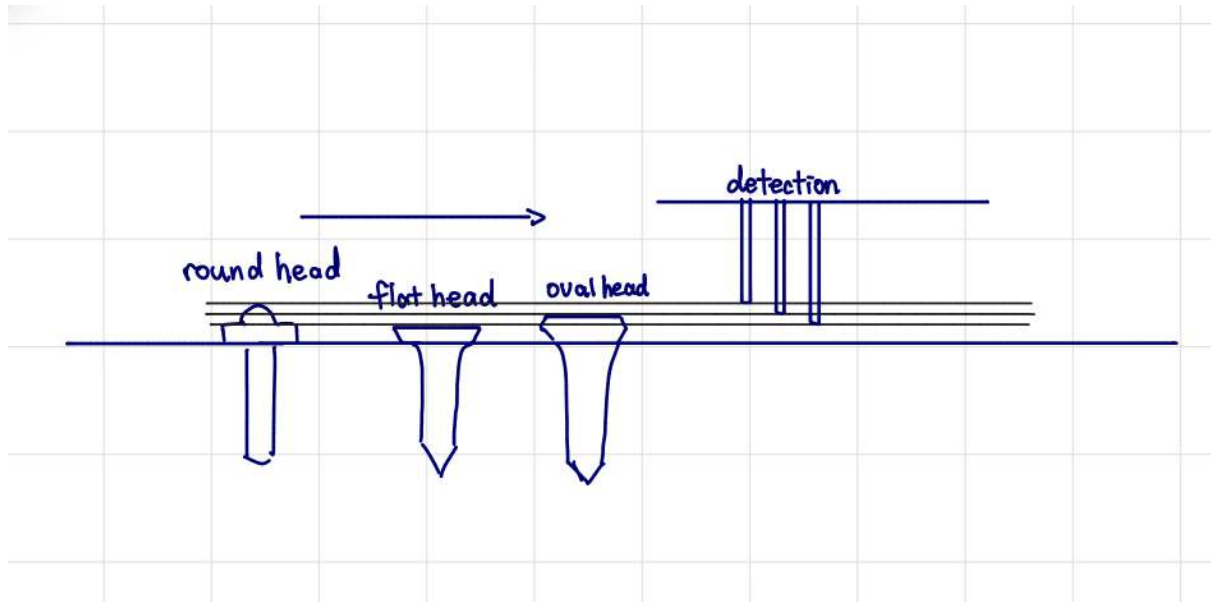


The vibration uses mechanical vibrations to move a pile of screws along a feeding track. When the bowl vibrates, friction between screws and the surface causes the screws to gradually move forward. By the end of the track, a small baffle makes sure they are all lined

up one by one in the same orientation horizontally, so that only the head can fit into the conveyor belt to move further, the screws that's not in the correct orientation will be pushed off the edge back into the bowl and wait to be bring up again.

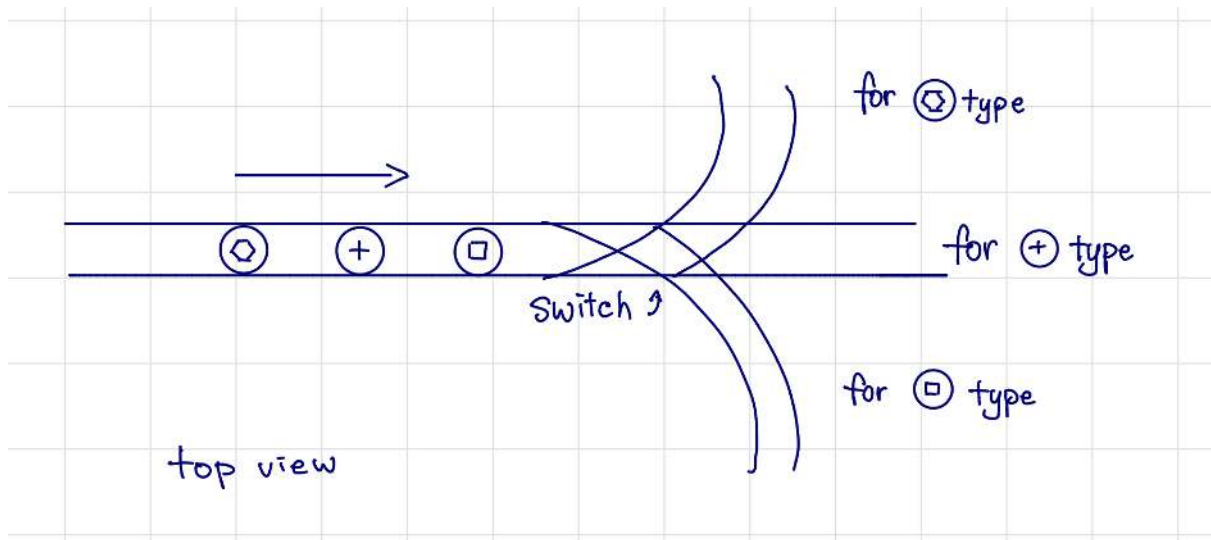
Graphic Prototype

1. Thickness of the head detection



The screws were aligned vertically along the conveyor belt, and were moved toward the detection stick. The stick will trigger the system once the screw head touches the corresponding length, and will be transported into the next layer of sorting.

2. Separate three different type of screw based on head shape into three different trails



After receiving the signal provided by the head detection system, the switch in the middle of the conveyer belt stick will switch the turnout so the screw can move to the right trail.

Pictures of imaging prototype:

